Che Hyun Lee

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EDUCATION

Seoul National University, Seoul, Korea

- B.S. in Electrical & Computer Engineering — Mar 2014 – Feb 2020
  - Focus: Machine Learning, Deep Learning
  - Cumulative GPA: 3.74 / 4.3 (Cum Laude)
- M.S. in Graduate School of Data Science — Mar 2020 – Feb 2022
  - Focus: Machine Learning, Deep Learning, Natural Language Processing, Data Science
  - Cumulative GPA: 4.1 / 4.3
- Ph.D. Candidate in Electrical & Computer Engineering — Mar 2022 –
  - Focus: Spoken Dialogue Systems, Diffusion Language Models, Spoken Language Models (SLMs), Diffusion-based Generative Models
  - Cumulative GPA: 3.9 / 4.3

RESEARCH INTERESTS

Spoken Dialogue Systems, Diffusion Language Models, Spoken Language Models (SLMs), Diffusion-based Generative Models

PUBLICATIONS

FIRST-AUTHOR PUBLICATIONS

[1] C. Lee, S. Park, D. Kang, D. Lee, Y. Cho, H. Kim, S. Yoon, "Phrase-Localized Language-Contrastive Guidance: Training-Free Localized Accent Control for Code-Switching Text-to-Speech," in ***EMNLP 2026, Oral Presentation***, Budapest, Hungary, Oct 2026.

[2] C. Lee, H. Kim, S. Yoon, "From Awareness to Adherence: Bridging the Context Gap in Spoken Dialogue Systems via Context-Aware Decoding," in ***INTERSPEECH 2026***, Sydney, Australia, Sep 2026.

[3] C. Lee, H. Kim, J. Yeom, S. Yoon, "EdiText: Controllable Coarse-to-Fine Text Editing with Diffusion Language Models," in ***ACL 2025***, Vienna, Austria, Jul 2025.

[4] H. Kim*, C. Lee*, S. Park, J. Yeom, N. Park, S. Yu, S. Yoon, "Does Your Voice Assistant Remember? Analyzing Conversational Context Recall and Utilization in Voice Interaction Models," in ***Findings of ACL 2025***, Vienna, Austria, Jul 2025.

CO-AUTHORED PUBLICATIONS

[5] D. Lee, S. Park, E. Song, C. Lee, Y. Cho, J. Kim, J. Yi, H. Kim, S. Yoon, "HEAR Who Said What: Unlocking Speaker-Attributed Reasoning via Counterfactual Voice Grounding," in ***EMNLP 2026***, Budapest, Hungary, Oct 2026.

[6] D. Lee, E. Song, C. Lee, H. Kim, S. Yoon, "Still Between Us? Evaluating and Improving Voice Assistant Robustness to Third-Party Interruptions," in ***ACL 2026***, San Diego, California, United States, Jul 2026.

[7] N. Park, C. Lee, J. Yeom, H. Kim, S. Yoon, "Beyond Language-Specific Neurons: The Challenge of Identifying Speech-Specific Neurons in Multimodal LLMs," ***IEEE Journal of Selected Topics in Signal Processing***, vol. 20, no. 1, Jan 2026.

[8] N. Park, H. Kim, C. Lee, J. Choi, J. Yeom, S. Yoon, "NanoVoice: Efficient Speaker-Adaptive Text-to-Speech for Multiple Speakers," in ***ICASSP 2025***, Hyderabad, India, Apr 2025.

[9] J. Yeom, H. Kim, J. Choi, C. Lee, N. Park, S. Yoon, "VoiceGuider: Enhancing Out-of-Domain Performance in Parameter-Efficient Speaker-Adaptive Text-to-Speech via Autoguidance," in ***ICASSP 2025***, Hyderabad, India, Apr 2025.

[10] H. Kim, S. Lee, J. Yeom, C. Lee, S. Kim, and S. Yoon, "VoiceTailor: Lightweight Plug-In Adapter for Diffusion-Based Personalized Text-to-Speech," in ***INTERSPEECH 2024***, Kos Island, Greece, Sep 2024.

[11] C. Shin*, H. Kim*, C. Lee, S. Lee, and S. Yoon, "Edit-A-Video: Single Video Editing with Object-Aware Consistency," in ***ACML 2023, Oral Presentation, Best Paper Award***, Istanbul, Turkey, Nov 2023.

*: Equal contribution

RESEARCH EXPERIENCE

Seoul National University, Seoul, Korea

- SNU-Samsung Electronics Co., Ltd. System LSI Collaboration Project — Dec 2023 –
  - Project: Streaming Personalized Text-to-Speech
  - Supervisor: Prof. Sungroh Yoon
  - Developed and trained causal diffusion TTS models for streaming synthesis, with chunk-level caching to reduce redundant computation.
  - Developed batch-wise LoRA adapters with parameter sharing for efficient multi-speaker personalization. Co-authored NanoVoice (ICASSP 2025).
  - Related work: Proposed training-free, localized accent guidance for code-switching TTS. First-authored Phrase-Localized LCG (EMNLP 2026, Oral).
  - Currently adapting the model for on-device inference and building an Android demo for streaming synthesis. Learned Android development with coding-agent assistance and implemented the required components without prior Android experience.
- SNU-Samsung Electronics Co., Ltd. DS Collaboration Project — Sep 2023 – Sep 2024
  - Project: Parameter-Efficient Fine-Tuning in Multimodal AI Models
  - Supervisor: Prof. Sungroh Yoon
  - Developed inference-time guidance to improve out-of-domain robustness in parameter-efficient TTS personalization. Co-authored VoiceGuider (ICASSP 2025).
- Undergraduate Research Student, Electrical & Computer Engineering — Mar 2019 – Aug 2020
  - Project: Prediction of CRISPR-Cpf1 with various ML/DL models
  - Supervisor: Prof. Sungroh Yoon
  - Implemented CNN-based prediction model for CRISPR-Cpf1 protein sequence data.

HONORS

Cum Laude, Seoul National University, 2020
Academic Performance Scholarship, Seoul National University: 2015-1, 2015-2, 2019-1

LANGUAGES

- Korean: Native language.
- English: Fluent (speaking, reading, writing).

REFERENCES

- **Professor Sungroh Yoon**
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